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ACME gas flow rate and velocity computations

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The ACME chamber insert features three crew-exchangeable Mass Flow Controllers (MFCs) where there is one each for fuel, oxidizer or inert, and nitrogen. An assortment of MFCs with different flow ranges is available for each of the three gas types. The number and range of MFCs for each gas type was based on operational plans and the ACME test matrices in the Integrated Science Requirements Document (ISRD).

The raw measurements for the gas flow rates are found with the rest of the non-imaging measurements in a comma-separated values (csv) file named for example 201235D1_data.csv. The test number is made up of the year (2-digts), day of year (3 digits), and test (a letter and number). As such, the example, 201235D1 is test D1 conducted on the 123rd day of 2020 or early on the 124th day because the day portion of ACME test numbers is not changed when operations continue past midnight (GMT) into the start of the next day.

The flow measurements are in the second, third, and fourth columns of the data file, for which the headings are in second row (while the first row includes a limited overview of the test conditions). By column, the flow rates are for:

B	MFCA1StaticMixerFlow	oxidizer or inert gas
C	MFCA2BypassFlow	nitrogen
D	MFCA3FuelFlow	fuel

These raw measurements are in voltages and can be converted to standard liters per minute (slpm) on a nitrogen basis per the conversions specified in the Software Interface Design Description (ACME-DOC-010 Rev. D), as seen in this excerpt from page 128:

10.2.2 MFCA1StaticMixerFlow
Description: Measured flow from MFCA1 sourced from the CIR static mixer manifold. This is the oxidizer.
Device: Teledyne Hastings HFC-D-302
Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA1 Full Scale
,where MFCA1 Full Scale is indicated in the meta data header.

10.2.3 MFCA2BypassFlow
Description: Measured flow from MFCA2 sourced from the CIR nitrogen or nitrogen bypass manifold. This is the inert.
Device: Teledyne Hastings HFC-D-302
Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA2 Full Scale
,where MFCA2 Full Scale is indicated in the meta data header.

10.2.4 MFCA3FuelFlow
Description: Measured flow from MFCA3 sourced from the CIR fuel manifold.
Device: Teledyne Hastings HFC-D-302
Conversion: Flow rate (SLPM) = Voltage / 5.0 * MFCA3 Full Scale
,where MFCA3 Full Scale is indicated in the meta data header.

The ACME MFCs are Hastings HFC-D-302 with default standard temperature and pressures (STP) of 0 Celsius and 760 torr for computations of volumetric flow or velocity. It is very important to understand that slpm is a measurement of the mass (or alternatively molar) flow rate where it is independent of the actual temperature or pressure. This is discussed for example in the MFC manual, where chapter 9 (on page 39) is "Volumetric vs. Mass Flow."

Conversions for the actual gases can be made with information provided in the manual on subsequent pages and the spreadsheets GasDataTables.xls and Inside gas mix calculator.xls which were both provided by the MFC manufacturer. First, open GasDataTables.xls and enable content, and then open Inside gas mix calculator.xls and enable content. From this, one should be able to enter mixtures and compute appropriate Gas Conversion Factors (GCFs).

Alternatively, the conversions above can be made with spreadsheets prepared by the ACME engineering team, where options include the following:

ACMEFlowCalculator.xlsx
ACMEFlowAndPressuresRawDataConversion.xlsx

Computation of gas velocities, for example for the gas-jet and coflow burners, depends on burner dimensions. For the coflow burner, ACME mechanical design engineer Adrian Drake indicated the following:

> "The inner diameter of the inner tube is .085 inches. Its cross sectional area is .00567 in^2, or 3.66 x 10^-6 m^2.
>
> The flow of the outer tube is a little tricky to calculate, based on where and how you measure. There's an internal step in the outer tube that can't be seen from outside, and blocks the flow to the outer honeycombs. Because of this, when we calculated the cross-sectional are of the outer tube, it measures from the inner step.
>
> The outer tube's outer diameter at that step is 0.984 inches. Its cross section area at that diameter is .7605 in^2. From this we have to subtract the inner tube's outer diameter: .095 inch diameter, so area is .00709. Subtract from the Outer Diameter and we get .7534 in^2, or .000486 m^2."

The dimensions of the coflow burner's inner tube are documented in greater detail in the ACME Inspection Verification Report (ACME-VER-001 Rev. A, ACME-VER-001_A.pdf) on pages 138-143 of the pdf file. From that section:

> The dimensions specified for this requirement are an I.D. of 0.0827 in. (2.1 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0850 in. (2.159 mm) for the I.D with minimum and maximum of the I.D., 0.0840 in. (2.134 mm) and 0.0860 in. (2.184 mm). The minimum and maximum values fall within the allowable tolerance range of 2.0-2.2 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the ID are met.
>
> The dimensions specified for this requirement are an O.D. of 0.0945 in. (2.4 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0950 in. (2.413 mm) for the O.D with minimum and maximum of the O.D. 0.0940 in. (2.388 mm) & 0.0960 in. (2.438 mm). The minimum and maximum values fall within the allowable tolerance range of 2.3-2.5 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the O.D. are met.

The 2.1-mm gas-jet burner used for the E-FIELD Flames experiment used the same tubing as the coflow burner's inner tube, where the tube dimensions are based on certification by the tubing manufacturer rather than measurement. The tubing dimensions for this burner are documented in greater detail in pages 102-104 of ACME-VER-001_A.pdf. From that section:

> The dimensions specified for this requirement are an I.D. of 0.0827 in. (2.1 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0850 in. (2.159 mm) for the I.D with minimum and maximum of the I.D., 0.0840 in. (2.134 mm) and 0.0860 in. (2.184 mm). The minimum and maximum values fall within the allowable tolerance range of 2.0-2.2 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the ID are met.
>
> The dimensions specified for this requirement are an O.D. of 0.0945 in. (2.4 mm). The tolerance is +/- 0.0039 in. (+/- 0.1 mm) and the accuracy is +/- 0.0020 in. (+/- 0.05 mm). Per the manufacturer's P/N listed on the drawing, the data sheet for this tubing has a nominal size of 0.0950 in. (2.413 mm) for the O.D with minimum and maximum of the O.D. 0.0940 in. (2.388 mm) and 0.0960 in. (2.438 mm). The minimum and maximum values fall within the allowable tolerance range of 2.3-2.5 mm. The tolerance of the Microgroup supplied tubing is +/- .05 mm. As the vendor tolerance is equal to the required precision of +/-.05 mm, both the tolerance and precision requirements on the O.D. are met.

Meanwhile, the circular outlet and housing of the two BRE burners are 25 and 50 mm in diameter, where those dimensions are documented on pages 158 and 187-189 of ACME-VER-001_A.pdf. From those pages:

> The BRE 25mm Burner Assembly (S1101MFA1245) features a machined Outlet Plate (S1101MFD1251). Per the drawing, the Outlet Plate's outer diameter is .984 +/- .005 (25 +/- .13mm). Per the vendor incoming inspection report (PO 36192) the flight parts vary in outer diameter from .984 to .9867. These values fall within the specified requirement. As the spread of measurements is less than required accuracy of +/- 0.1mm, both parts of this requirement are verified.
>
> The BRE 25mm Burner Assembly (S1101MFA1245) features a top housing (S1101MFD1250) that interfaces with the burner outlet. The top housing has an inner wall dimension on the drawing of .987-.989 (25.07mm-25.12mm). Per the vendor incoming inspection (PO 36197), the flight hardware measures .987 (25.07mm), which falls within the range of the requirement (24mm – 26mm). This requirement is verified.
>
> 36192
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> The BRE 50mm Burner Assembly (S1101MFA1230) features a machined Outlet Plate (S1101MFD1237). Per the drawing, the Outlet Plate's outer diameter is 1.969 +/- .005 (50 +/- .13mm). Per the vendor incoming inspection report (PO 36192) the flight were all made to the same dimension of 1.969. These values fall within the specified requirement. As the spread of measurements is less than required accuracy of +/- 0.1mm, both parts of this requirement are verified.
>
> The BRE 50mm Burner Assembly (S1101MFA1230) features a top housing (S1101MFD1236) that interfaces with the burner outlet. The inner wall of the BRE burner is a part of the top housing. Per the drawing, the inner wall dimension is 1.972-1.974 (50.09-50.14mm). Per the vendor incoming inspection (PO 36192) the flight parts were all made with an inner wall dimension of 1.972. This requirement is verified.

Of course, dimensions are also available from the burner drawings, but the dimensions are often not found on the assembly drawings so the appropriate component drawings must typically be consulted.

Commercial products are identified for reference and are not meant to indicate endorsement by NASA or the federal government.